



AI HACKATHON

EMBRACING THE FUTURE

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INITIATIVE NAME

❑ **Category Name:** Education -AI Driven Personalized Solution

❑ **Group Name:** Buraq

❑ **Project Name:** QASIM (Quality Assessment of Simulation)

❑ **Team Members:**

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- III. Md. Morakib Hossain
- IV. Mir Oliul Pasha Taj

PROBLEM STATEMENT

Solving engineering mathematical problems like complex hand-drawn electrical circuits can be confusing and time consuming, which we intend to solve.

Challenges in Circuit Solving:

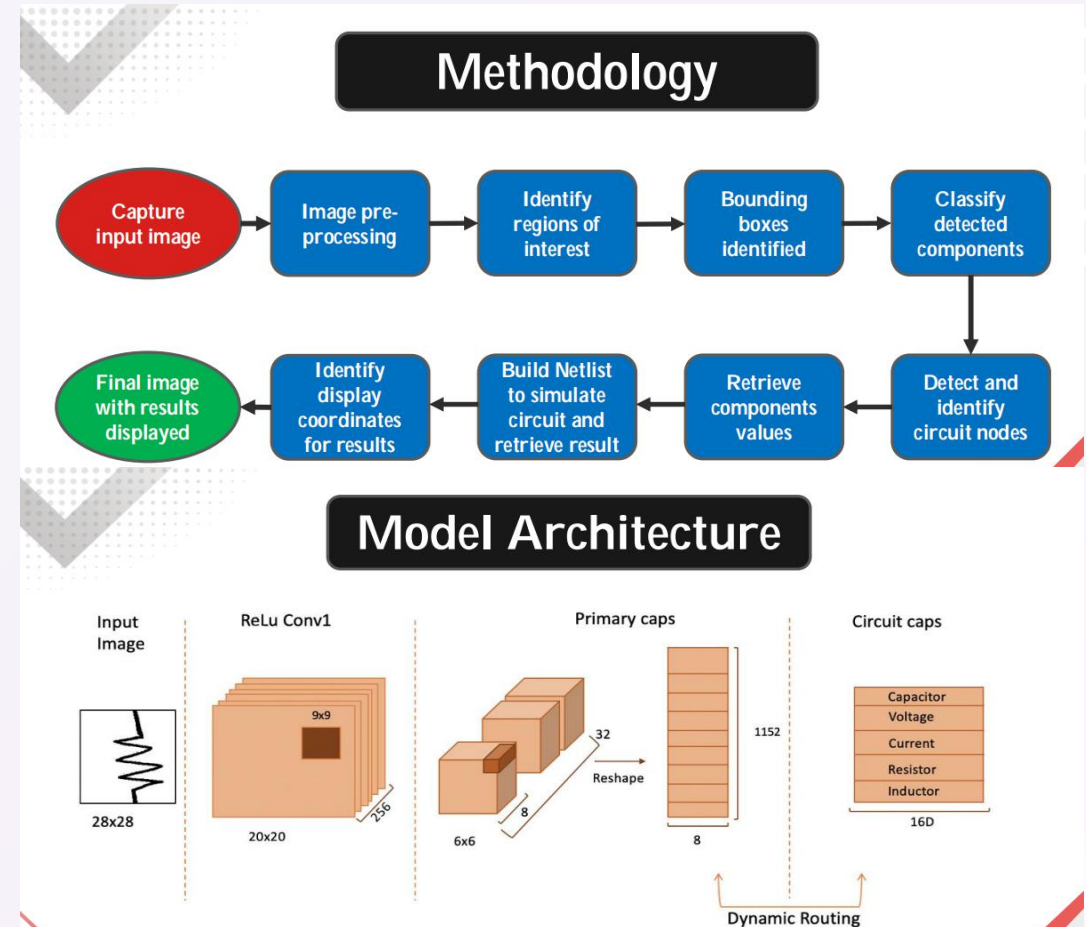
- Existing circuit simulators require installations and workstations.
- Students face difficulty in recognizing hand-drawn circuits.
- Traditional tools demand extensive training data for accuracy.



AI-BASED SOLUTION BRIEF

Uses **Capsule Networks** to recognize and interpret hand-drawn circuits.

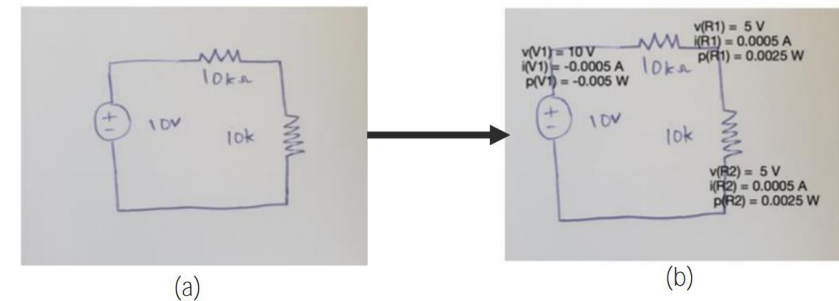
- Integrates **Augmented Reality (AR)** for real-time feedback.
- Requires **less training data** while improving recognition accuracy.
- **Mobile-based solution** to make circuit analysis accessible anytime, anywhere



EXPECTED IMPACTS

- Integration of machine learning techniques, particularly capsule networks, to recognize and detect circuit elements from images captured by smartphones.
- Introduction of capsule networks as a novel technique in deep learning to significantly reduce the training data requirement for circuit recognition.
- Development of a smartphone application for real-time analysis of hand-drawn circuits.

Results



Performance analysis

	Precision	Recall	F1-Score
Capacitor	1.00	1.00	1.00
Current	0.92	0.91	0.91
Resistor	0.99	1.00	1.00
Voltage	0.92	0.89	0.90
Inductor	0.97	1.00	0.99
Micro Avg.	0.96	0.96	0.96
Macro Avg.	0.96	0.96	0.96
Weighted Avg.	0.96	0.96	0.96

RESPONSIBLE AI USAGE STATEMENT

- **Ethical Considerations:** In this segment there AI can only support mankind.
- **Limitations:** Our system currently can solve electrical circuits only but we are working on adding different sectors.
- **Bias and Accuracy:** System is being trained to generalize different handwritten styles.

References:

- ❖ [1] Rehmat, A.; Hartley, K. Building Engineering Awareness: Problem Based Learning Approach for STEM Integration. Interdiscip. J. Probl.-Based Learn. 2020, 14, n1.
- ❖ [2] Sabour, S.; Frosst, N.; Hinton, G.E. Dynamic Routing Between Capsules. In Advances in Neural Information Processing Systems Curran Associates, Inc.: NewYork, NY, USA, 2017; pp. 3859–3869.
- ❖ [3] Rabbani, M.; Khoshkangini, R.; Nagendraswamy, H.; Conti, M. Hand Drawn Optical Circuit Recognition. Procedia Comput. Sci. 2016, 84, 41–48.

The background features a vibrant, abstract design. It includes a blue-to-purple gradient bar at the top, a light gray central area, and a purple-to-magenta gradient bar at the bottom. Decorative elements include a grid of white dots in the top right, a cluster of white dots in the bottom left, and various colorful, semi-transparent geometric shapes like triangles and circles scattered throughout.

Thank you